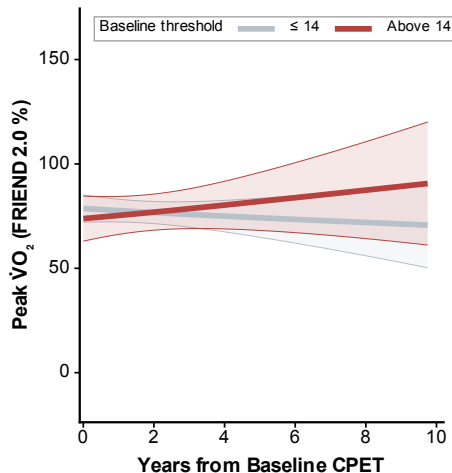


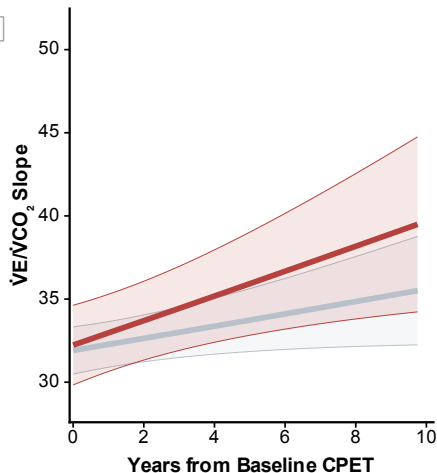
### (A) Peak $\dot{V}O_2$

Baseline  $E/e' \leq 14$  versus above 14



### (B) $\dot{V}E/\dot{V}CO_2$ slope

Baseline  $E/e' \leq 14$  versus above 14



### (C) Longitudinal Model Summary

| Model component                                | edf  | F statistic | P value |
|------------------------------------------------|------|-------------|---------|
| <b>Peak <math>\dot{V}O_2</math></b>            |      |             |         |
| Time smooth                                    | 1.06 | 0.55        | 0.493   |
| Threshold group                                | —    | 0.00        | 0.975   |
| Time x threshold group                         | 1.01 | 1.38        | 0.240   |
| <b><math>\dot{V}E/\dot{V}CO_2</math> slope</b> |      |             |         |
| Time smooth                                    | 1.04 | 4.47        | 0.033   |
| Threshold group                                | —    | 0.62        | 0.431   |
| Time x threshold group                         | 1.03 | 1.31        | 0.249   |

**Supplemental Figure.** Parameter-specific longitudinal cardiopulmonary trajectories stratified by binary baseline  $E/e'$  (average) status. **(A)** GAMM-derived Peak  $\dot{V}O_2$  trajectories. **(B)** GAMM-derived  $\dot{V}E/\dot{V}CO_2$  slope trajectories. **(C)** Summary of the key status and interaction terms for both models. Baseline threshold groups were defined as  $E/e' \leq 14$  versus above 14. Longitudinal models were adjusted for age, sex, and BMI.